

Customer Sentiment on Twitter

Executive AI Analysis: From Conversation Data to
Decision Signal

The Objective: Turn Conversations Into a Decision Signal

Customer-support tweets contain direct signals of friction, uncertainty, and resolution. The project demonstrates how an AI-assisted workflow can convert raw conversations into a structured management view while retaining human control over meaning and action.

Management question	Analytical response
What is the customer saying?	Score inbound customer language for sentiment.
Where might friction concentrate?	Review issue themes, support context, and negative signals.
What can be automated safely?	Dataset orientation, exploratory visualisation, and scalable classification.
What must remain human-owned?	Interpretation, escalation, root cause, and decisions.

A Three-Tool Workflow Balances Speed and Control

The workflow moves deliberately from orientation to exploration to classification. Each tool contributes productivity while final interpretation remains analyst-led.

Stage	Tool	Output	Governance checkpoint
1. Orient	Gemini	Schema narrative, missingness, hypotheses	Confirm facts against the computed profile.
2. Explore	AutoViz	Rapid distributions and pattern candidates	Reject small-cell or non-material patterns.
3. Classify	Twitter-RoBERTa	Negative, neutral, positive customer signals	Review confidence, sarcasm, and context.

The Supplied Sample Proves the Workflow, Not Performance

The 10–12 October 2017 observation window validates the pipeline and demonstrates interpretation. It does not support enterprise-wide performance or brand rankings.

93

total observed tweets

49

inbound customer tweets

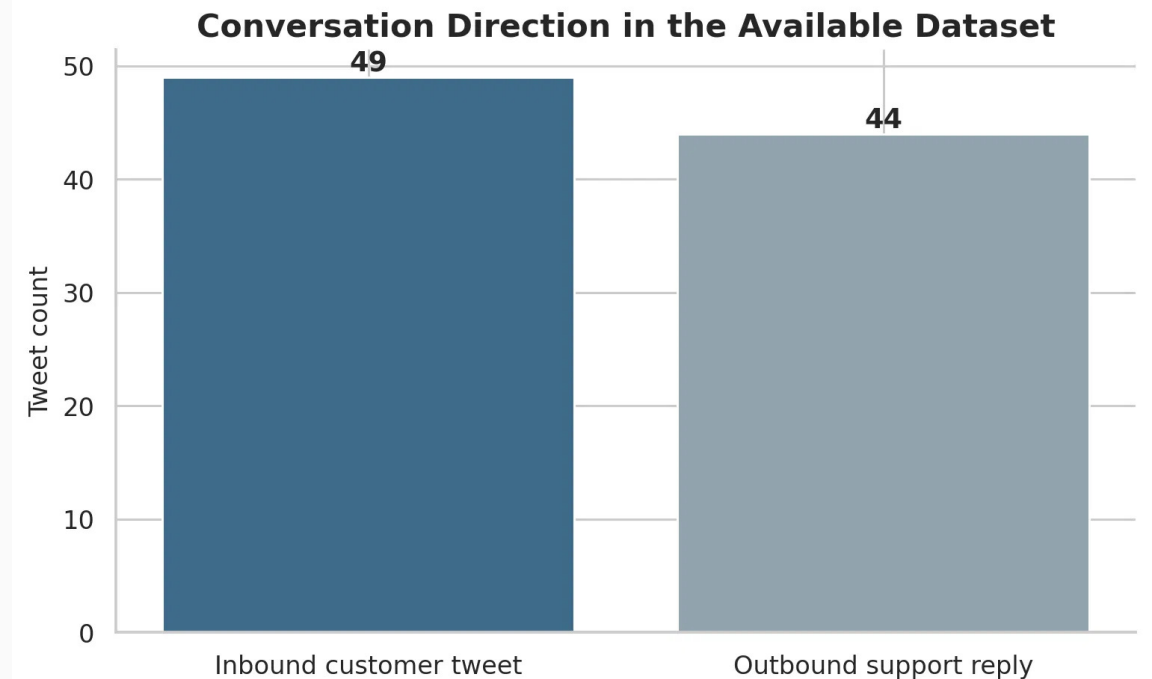
44

outbound support replies

13

support accounts

Every insight is explicitly scoped to this observed sample.



The Observed Customer Voice Is Friction-Led

Among 49 inbound customer tweets, the Twitter-specific model classified 53.1% as negative. This is a triage signal, not a customer-satisfaction score.

26

Negative · 53.1%

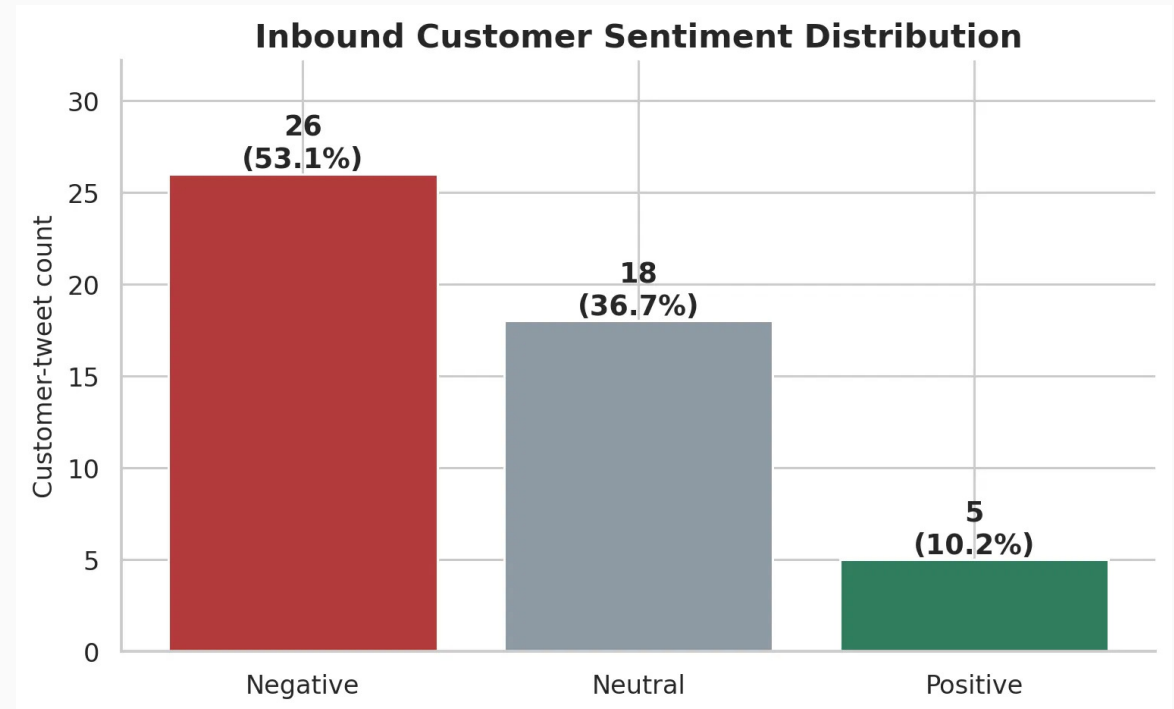
18

Neutral · 36.7%

5

Positive · 10.2%

**Prioritise the negative cohort
for context-aware issue review
and operational linkage.**



Product and App Performance Is the Strongest Named Signal

The transparent keyword taxonomy makes product and app performance the largest named theme. The unclassified bucket is a taxonomy-improvement opportunity, not an issue category to ignore.

Product & app performance

12 inbound tweets · 9 negative

Connectivity & service outage

5 inbound tweets · 3 negative

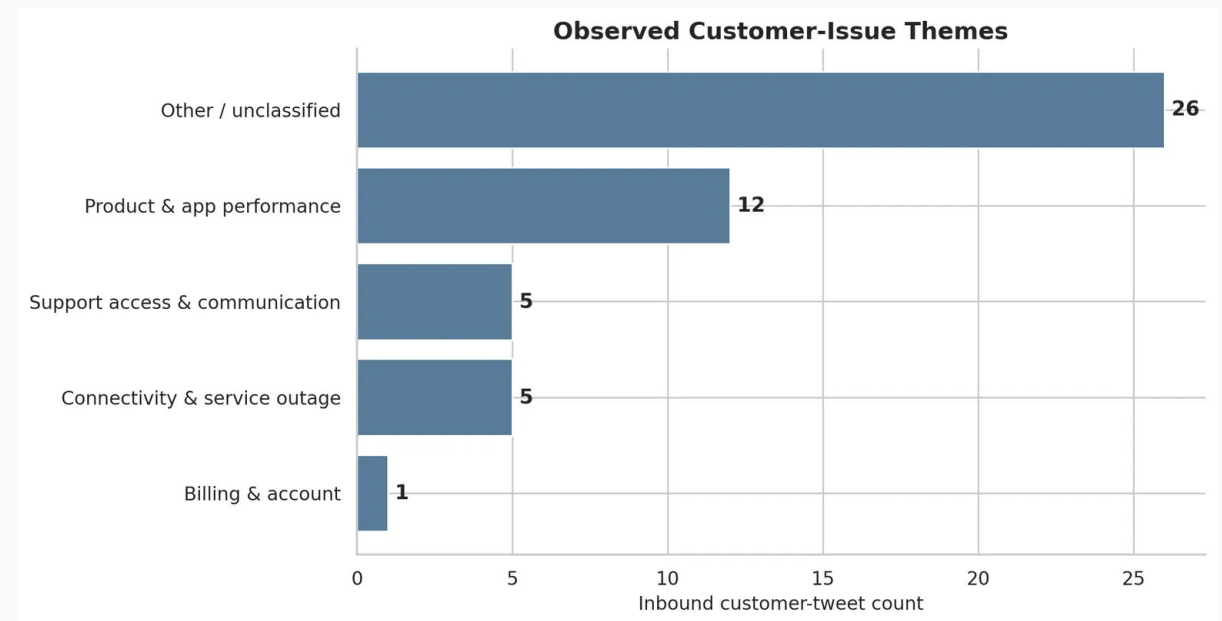
Support access & communication

5 inbound tweets · 3 negative

Other / unclassified

26 inbound tweets · coverage gap

Investigate update, app, and performance language first; improve the taxonomy in parallel.



Context Converts Sentiment Into an Actionable Signal

Sentiment is only one layer. Conversation direction, timestamps, reply links, and support-account context distinguish a complaint, a diagnostic exchange, and a confirmation of resolution.

Inbound vs. outbound	Separates the customer voice from the support response.
Reply relationships	Reconstructs the exchange around a sentiment signal.
Timestamp	Enables trend, incident, and product-release correlation.
Account attribution	Supports investigation with minimum-volume safeguards.

Management implication: link language signals to response time, resolution, incidents, and releases before making decisions.

AI Delivered Speed, Not Substitute Judgment

Gemini reduced initial orientation effort, AutoViz surfaced candidate patterns, and Twitter-RoBERTa scaled sentiment labelling. The analyst selected decision-relevant evidence and set boundaries on interpretation.

Tool	Productivity gain	Human decision required
Gemini	Converts metadata into a first-pass explanation	Validate facts, risks, and analytical priorities.
AutoViz	Produces exploratory visuals rapidly	Select material patterns and remove noise.
Twitter-RoBERTa	Scores customer language at scale	Validate ambiguity, sarcasm, and business relevance.

Executive implication: AI shortens the path to focused questions. It does not own the answer.

A Twitter-Specific Model Improves Domain Alignment

The selected Cardiff NLP RoBERTa-base model was trained on approximately 124 million tweets and fine-tuned for TweetEval sentiment classification. It is better aligned to Twitter language than a generic sentiment default.

Modelling choice	Rationale
Inbound tweets only	Captures customer sentiment rather than the company voice.
Three sentiment classes	Supports negative, neutral, and positive language signals.
Handle and URL normalisation	Reduces irrelevant token variation.
Deterministic sampling cap	Keeps large-corpus inference reproducible and runtime-safe.

Source: Cardiff NLP Twitter-RoBERTa sentiment model card.

Sarcasm and Low Confidence Need a Human Review Loop

No human sentiment labels are supplied, so accuracy cannot be reported honestly. The workflow exposes uncertainty and creates a review queue for controlled validation.

Validation signal	Observed result
Mean model confidence	0.787
Median model confidence	0.833
Tweets below 0.60 confidence	7 of 49 (14.3%)
Stratified audit records	12
Ground-truth accuracy	Not available

A clearly sarcastic message received a Positive label despite product-failure language. Context must override text-only polarity when actions are taken.

A Management-Ready Operating Model Starts With Triage

The prototype can mature into a governed monitoring capability by combining AI signals with operational evidence and clear accountability.

Time horizon	Priority action	Outcome
0–30 days	Review negative and low-confidence tweets; define minimum-volume rules.	Controlled triage and reliable escalation.
31–60 days	Add issue taxonomy, response-time, incident, and resolution data.	Contextual root-cause investigation.
61–90 days	Establish monitoring baseline, reviewer calibration, and governance KPIs.	Sustainable decision support.

Recommendation: start with a human-reviewed negative-sentiment queue, not a public scorecard.

From Raw Tweets to Responsible Insight

AI accelerates exploration.

Human judgment turns evidence into action.

Sources and Method Notes

Reference	Use in the project
Kaggle: Customer Support on Twitter	Dataset source and column structure.
Google AI: Gemini API libraries	Gemini SDK setup for optional initial exploration.
AutoViML: AutoViz	Automated exploratory visualisation and row-capping guidance.
Cardiff NLP: Twitter-RoBERTa sentiment model	Domain-specific sentiment model and label interpretation.

Full URLs, citations, reproducibility notes, and caveats are available in the GitHub README and Colab notebook.

Questions?

Thank you for your time. I welcome any questions or discussion on how we can operationalize these insights.

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